

Everything Is a Relationship

The Nonegalitarian, Log-Scaled Brain

We only use ten percent of our brain¹

You have no idea, how much poetry there is in the calculation of a table of logarithms.

—KARL FRIEDRICH GAUSS²

For things to stay the same, everything must change.

—GIUSEPPE DI LAMPEDUSA³

My fellow New Yorkers are tolerant people with a “we have seen it all” attitude. You may wear anything you want or nothing while walking the avenues. You can be a cross-dresser, or even have a Hungarian accent, and the majority of New Yorkers would not even blink. But

1. The exact origin of this neuromyth (BrainFacts.org) is not known. “In fact, most of us use only about 10 percent of our brains, if that.” wrote the Hungarian-Jewish illusionist, magician Uri Geller in 1996. [NeuroNote: Geller suffered from bulimia and anorexia nervosa. He believed he had paranormal powers given to him by extraterrestrial agents.] The US Defense’s Advanced Research Projects Agency (DARPA) wasted tons of dollars to examine the outrageous claims of this parapsychologist (Targ and Puthoff, 1974). The 10% idea may have been derived from a William James quote, “We are making use of only a small part of our possible mental and physical resources” (James, 1907; p. 12). Of course, 10% could refer to brain volume, firing rates, synaptic strengths, etc.

2. <http://lymcanada.org/the-poetry-of-logarithms/>.

3. di Lampedusa (1958).

a squirrel-sized man walking with a woman the size of a giraffe in the foyer of the Metropolitan Opera would surely raise some eyebrows, not because it is deemed scandalous but because it is so improbable.⁴ Such big variation in body size within a species is outside the norm. But in the brain, distributions that range over several orders of magnitude are the norm, whether we look at connectivity or dynamics. Magnitude or fold differences are easily noticeable because they are based on a ratio or relationship judgment. As explained in this chapter and the next, ratio judgment is a natural function of the brain due to the way it calculates.

Multiplication, division, fraction, proportion, normalization, and gain have been recurring terms in the preceding chapters⁵ because they are ubiquitous operations in brain circuits. All these operations utilize ratios, or *logos* in Latin. Hence the theme of this chapter: log-scaling in the brain.⁶ My goal here is to demonstrate that brain connectivity is far from random and that the brain's plastic features do not conform to the assumed substrate of a *tabula rasa* brain. Instead, a highly structured connection matrix supports a skewed dynamic that constrains how neuronal groups respond and maintain self-organized activity. To perform effectively, physiological brain operations must occupy a

4. While this scenario is highly improbable in practice, it is not theoretically impossible. Domesticated dogs have the highest body size variation of any species, from the 5-inch tall Chihuahua to the 7-foot tall Great Dane, thanks to human breeders. Body size and brain size variations show a normal or bell-shaped (Gaussian) distribution in all natural species.

5. The relationship between addition and multiplication has been discussed in various cultures. In medieval Europe, Christian scholars claimed that the knowledge-based road to God was a product of learning the scriptures of the Bible and using logic. The argument in favor of multiplication was that no amount of logic will help without reading the scriptures. If scripture knowledge is zero, the total knowledge will be still zero. Conversely, if your logic value is zero, then your knowledge will be still zero even if you have memorized every page in the Bible (Harari, 2017). By a similar logic, yin and yang are inseparable and opposite forces according to Chinese philosophy. No amount of yin or yang alone means knowledge or balance. In contrast, addition would always yield some knowledge even if you do not read the Bible or you are ignorant of the yang.

6. *Logos* has multiple meanings, but in the mathematical context, it refers to *ratio* or *proportion*. "Logarithm" translates to numbers with ratios, as *arithmos* means number. The problem of logarithm evolved from astronomical observations that had to address ratios in spherical trigonometry, such as the sine, cosine, tangent, and cotangent of angles. In connection with logarithms, the Scot John Napier deserves to be remembered. Napier's main contribution was to introduce a practical device (the tables of logarithms). He recognized that multiplication or division of two positive variables can be calculated by adding or subtracting the logarithms and taking the antilog of that sum or difference, respectively. Unknown to Napier, the Swiss watchmaker Jost Bürgi had already prepared such log tables, which were used by Johannes Kepler (Clegg, 2016). [NeuroNote: Kepler's mother was accused of practicing witchcraft, and several of his family members had mental problems.]

wide dynamic range between silence and supersynchrony. This dynamic range is needed to allow many competing features to coexist in the brain, such as redundancy, resilience, degeneracy, homeostasis, plasticity, evolvability, robustness, and stability. These features constantly compete with each other.⁷ In simpler terms, networks must be responsive to both weak and strong inputs. Weak stimuli should be amplified, whereas strong stimuli should be attenuated. At the same time, neuronal circuits should maintain their ability to evaluate differences in inputs. That is, networks must be both sensitive and stable. If these competing demands do not strike the right balance, the network becomes hyperexcitable and epileptic or remains dead silent forever. The delicate balance between these competing dynamic features is reached via components that are diverse, show a large quantitative variation, and combine in multiplicative ways. Systems with such properties are energy efficient and show high tolerance to component failures.⁸ How to build and maintain such systems?

DIVERSITY YIELDS INTERESTING SYSTEMS PROPERTIES

Attending the annual Society for Neuroscience meeting always makes me feel anxious. That feeling arises from a specific uncertainty: What difference would it make if I were not part of this enterprise? In this stock exchange of ideas, 25,000 of my hard-working colleagues present their results. All the presentations contain something worth knowing, many of them are interesting, and there are always a handful that I cannot miss. The really interesting discoveries often come from the same top laboratories, year after year. I would guess that half of neuroscience knowledge comes from 10–20% of the labs. However, that also means

7. Some of these terms may not be clear to most readers. *Robustness* is the persistence of a function under perturbations. The somewhat related *resilience* describes the ability of a system to recover after challenges. *Redundancy* refers to components whose presence are not critical for a given function unless other components fail, in which case their role becomes critical (von Neumann, 1956). It is like having a chain on the door in addition to the lock. *Degeneracy* says that the same function can be achieved by a variety of different mechanisms. For example, a rhythm of the same frequency can be generated by a variety of structural-functional solutions. *Homeostasis* is a system's ability to maintain things (e.g., balance of inhibition and excitation through oscillations or other solutions). *Evolvability* is the capacity of a system to adapt to novel situations. One can call it *learning*. Learning, homeostasis, and adaptation to perturbation are achieved through plasticity.

8. Mitra et al. (2012); Ikegaya et al. (2013); Helbing (2013). An advantage of logarithmically linear computation in both electronic circuits and biochemical systems is its energy efficiency (Sarpeshkar, 2010; Daniel et al., 2013). This principle likely applies to brain circuits as well.

that the other half comes from the remaining 80–90% of investigators. The two halves of the distribution cannot function efficiently without each other.⁹ Here is why.

Division of labor, made possible by diversity, is perhaps the most ubiquitous rule in living systems. The brain is not an exception. It contains many different neuron types and non-neuronal cells. In complex systems, this is known as *component diversity*. As an analogy, electronic circuits have a variety of different components, such as transistors, diodes, resistors, and capacitors. However, the proud transistor with its nonlinear amplification abilities cannot shine without the modest but critical contribution of the humble resistor. Even within components, there is a great variance. For example, the gain of a transistor can be small or large, and the resistance of resistors can vary by several orders of magnitude. Similarly, in the brain, the quantitative features within the same neuron types, synapses, and networks vary by ten- to a hundredfold and occasionally a thousandfold.

When many identical components interact with each other additively, we can get a good idea about their average behavior from macroscopic observations. For example, the properties of gases can be described by simply measuring their overall temperature without knowing the position and velocity of every molecule. As the number of similar components increases, the solutions typically become easier to compute.¹⁰ In contrast, describing the precise behavior of one atom remains a huge challenge for physics. Yet the toughest realm to evaluate is when dozens of qualitatively different components interact. This is exactly the case in neuroscience. An analogy would be an orchestra, where the interactions of different instruments give rise to a virtually limitless variety of different compositions.

In an invertebrate neural circuit, such as the somatogastric ganglion of the crab, there are only a handful of neurons, yet each of them is different. Although the entire “wiring diagram” of this small circuit has been determined, we do not know the exact mechanisms through which these cell-to-cell interactions give rise to the sequential spatiotemporal patterns that produce a rhythmic

9. You may ask how such an allegedly skewed distribution of creativity relates to the normal distribution of the intelligence coefficient (originally *Intelligenzquotient* of William Stern; IQ), known to be a perfect bell curve (Herrnstein and Charles, 1994). The IQ curves are made symmetric by designing the tests to meet specific criteria. It does not measure creativity but aptitude to fit a particular niche.

10. This approach works because molecules of the same type are indistinguishable. We cannot label them and recognize them individually later. Neurons are different. They are individuals like we are. In fact, each neuron carries a distinguishable “barcode” that allows investigators to track its fate from birth to end of life (Kebschull et al., 2016; Mayer et al., 2016).

output.¹¹ Similarly, small brains have relatively few neurons compared to large brains, but their component diversity is high. However, component diversity grows only modestly across species.¹² Instead, a few neuron types, such as pyramidal cells, multiply in great numbers in larger brains, in a process akin to adding a million second violinists to an orchestra. The good news is that a system built from ten types of neurons, with one type multiplied a millionfold, is not vastly more challenging to understand than a tiny brain with just ten different individual neurons.¹³ The challenge is still huge (though not hopeless) because, even within the same type of component, quantitative variations are very large, as will be discussed later. Let us start the discussion on skewed distributions with an old problem in cognition and attempt to assign a neuronal mechanism to it.

THE LOG RULE OF PERCEPTION

Modern neuroscience, like early physics, is striving to describe and condense observations with mathematical equations. Compared to physics, neuroscience has only a modest number of laws.¹⁴ Yet, one of these, the *Weber law* or also called the *Weber-Fechner law* has a breathtaking simplicity and generality. This law is named after two German scientists who laid the groundwork for human psychophysics, a quantitative investigation of the relationship between physical stimuli and the mental states they induce. Ernst Heinrich Weber was

11. Harris-Warrick et al. (1992); Marder et al. (2015). The degeneracy of this circuit is very rich, with hundreds of circuit solutions to the same rhythm in related species and within the members of the same species (Prinz et al., 2004).

12. For example, GABAergic interneuron types appear to be preserved in mammals (Klausberger and Somogyi, 2008). Principal cells of different cortical layers in different species also show strong similarities. Every now and then, novel neuron types, such as “spindle cells” in bonobos and whales are reported (Chapter 9), but they seem to be the exception rather than the rule.

13. A similar sentiment has been expressed by Gao and Ganguli (2015), who suggest that the number of neurons that need to be recorded in an experiment is proportional to the logarithm of the neuronal task complexity, although the role of component diversity was not explicitly addressed in this work.

14. Of course, there are no laws and rules in nature. In the original use of the term by Isaac Newton, a law refers to an event governed or caused by an external agent. According to Newton, all laws are enforced by God and never change. In reality, there are only reliable regularities that appear to obey some fictional laws, such as the law of gravity, Ohm’s law, or the psychophysical Weber-Fechner law. These reliable regularities and recurrences of the world are exploited by the brain to predict the future.

a physician who was interested in how we can perceive differences in touch. After extensive experimentation, he concluded that “in observing the disparity between things that are compared, we perceive not the difference between the things, but the ratio of this difference to the magnitude of things compared.” For example, if you are holding a 100-gram object, the second object needs to weigh at least 110 grams in order for you to notice a difference. If the object weighs 200 grams, you can detect a weight difference only if the other object weighs more than 220 grams or less than 180 grams. This threshold change, called the *Weber fraction*, is also called the *just noticeable difference*.¹⁵

Gustav Theodor Fechner did not conduct experiments. Instead, we can call him an early computational scientist. He trusted Weber’s observations and calculated mathematically that sensation is a logarithmic function of physical intensity. Therefore, when stimulus strength multiplies, the strength of perception adds.¹⁶ If the importance of a law depends on its generalizability, the Weber-Fechner rule is important. It applies to vision, hearing, and taste. Distance perception, time perception, and reaction time also vary logarithmically with the distance or time interval, respectively.¹⁷ The distribution of intersaccadic intervals—that is, how long we inspect a visual scene—also shows a log-normal form. Similarly, it takes an increasingly longer time to discriminate between two numbers as the difference between them decreases. Decision-making and short-term memory error accumulation also obey the law.¹⁸ This is an impressive list.¹⁹ The Weber-Fechner law was conceived 150 years ago, but the reason that our subjective sense of many variables contains these particular systematic patterns has remained obscure.

15. The just noticeable difference can be considered as a threshold for subjective judgment.

16. “Simple differential sensitivity is inversely proportional to the size of the components of the difference; relative differential sensitivity remains the same regardless of size” (Fechner, 1860/1966). $S = k \ln I + C$, where S stands for sensation, I for stimulus intensity, C is the constant of integration, and $\ln$ is the natural logarithm. The constant k is sense-specific and varies across modalities. Fechner claimed that his insight into the relationship between material sensation and mental perception came to him while lying in bed half asleep (Heidelberger, 2004).

17. When we are asked to estimate the length of an interval, the variability in our responses scales with the magnitude of the interval to be timed (Gibbon, 1977; Gibbon et al., 1984; Wearden and Lejeune, 2008). Similarly, errors of distance estimation increase proportionally with the distance to be estimated.

18. Dehaene et al. (1998); Gold and Shadlen (2000); Deco and Rolls (2006); Buzsáki (2015). Stevens (1961) proposed that a power-law relation between sensation and stimulus intensity is more mathematically plausible than the log rule. However, Stevens’s approach has been repeatedly criticized on methodological grounds (Mackay, 1963; Staddon, 1978).

19. Many nonbiological phenomena, such as sound level, earthquake intensity, pH scale, and entropy, also exhibit logarithmic behavior.

My hypothesis to explain these systematic patterns is based on the assumption that the subjective properties of our actions and perceptions are supported by the log-normal distribution of the meso- and microscopic “connectome” of the brain and its log rule–dominated dynamics.²⁰ To support this hypothesis, we must find a matrix of connections and a dynamic in the brain that match the statistical features found in psychophysics experiments. I agree that this explanation is not the only possibility, but it is worth pursuing. Before I can explain this idea further, we need to discuss the nature of some statistical distributions.

NORMAL AND SKEWED DISTRIBUTIONS

No person has a brain ten times larger than that of an average human. Brain volume within the same species shows a normal or Gaussian distribution.²¹ A *normal distribution* is characterized by a bell-shaped curve that is symmetrical and can be quantified by two parameters—the mean and the standard deviation. In a true normally distributed population, the incidence of values several times larger at the left hand compared to the right hand is practically zero.

However, normal distributions are rare in biology. Driven by diversity, large variations are the norm rather than the exception in biological systems. Multiplication, rather than addition, of proportionate values gives rise to many “extreme” values, which make the distribution look asymmetric or “skewed.” Such distributions can take many forms.²² Perhaps the most common skewed

20. This idea was the major theme of my previous book (*Rhythms of the Brain*, 2006). Similarly, the Bayesian model of the brain is a quantitative attempt to describe the match between world events and constructions of patterns by the brain (Helmholtz, 1866/1962; Ashby, 1947, 1962; Dayan et al., 1995; Rao and Ballard, 1999; Friston, 2010; Friston and Buzsáki, 2016).

21. Physical quantities that are the sum of many independent variables often have normal distributions. According to the Central Limit Theorem from probability theory, the addition or subtraction of a large number of small independent random variables yields a frequency distribution that takes the form of a Gaussian curve. Not so long ago, scientists believed that all distributions in nature are normal, which is why it was called “normal.” The idea was that there is a typical value, and the variability is due to some additive noise. Perhaps Henri Poincare was the first to question this belief: “Everyone believes in it: experimentalists believing that it is a mathematical theorem, mathematicians believing that it is an empirical fact” (as cited in Lyon, 2014).

22. Gamma distribution and log-normal distributions are similarly skewed, but gamma does not strictly rise from the multiplication and division of random variables. For log-normal, the skewness of a log plot is zero, whereas a gamma distribution has a heavier tail on the left. The fit of the different distributions to the data can provide clues to the generative mechanisms that give rise to the distribution.

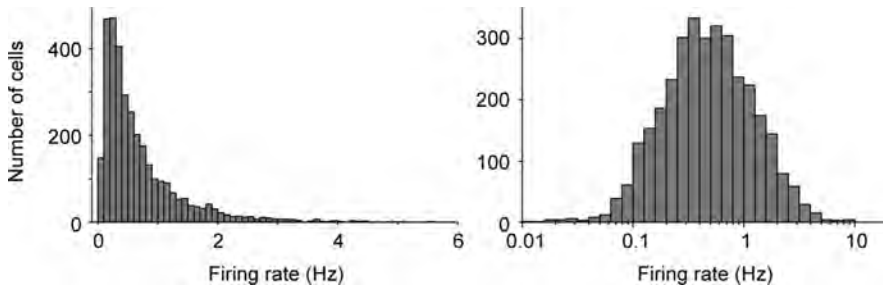


Figure 12.1. Relationship between linear and log display of a distribution. Firing rate of hippocampal pyramidal neurons during non-rapid eye movement (non-REM) sleep, plotted in the linear scale. Log-normal distribution is unimodal on the log scale.

distribution in biology is the logarithmic-normal or log-normal distribution.²³ This distribution is right-skewed on a linear scale but looks bell-shaped when the logarithms of the observed values are plotted. In other words, a log-normal distribution is a probability distribution of a random variable whose logarithm is normally distributed. Examples from biology include the number of species per family, survival times of species, sizes of fruits, pharmacological effects, times to first symptoms of infectious diseases, and blood pressure distribution within an age group, just to list a few. Basic physiologic processes, such as resting heart rate, visual acuity, and metabolic rate, also show a skewed distribution. In economics, incomes are often well fitted to a log-normal function.²⁴

The log-normal distribution also describes the firing rates of cortical pyramidal neurons. The activity of these cells varies from nearly complete silence to several spikes per second (Figure 12.1). When the long-term average firing rates of many neurons are plotted on a linear scale, the distribution is strongly skewed, with many slowly firing neurons at the left end and a minority of highly

23. Aitchison and Brown (1957); Crow and Shimizu (1988); Ansell and Phillips (1994); Limpert et al. (2001).

24. Log-normal distributions describe data from diverse disciplines of science (Koch, 1966). Many of these distributions in biology have also been characterized by power laws. In a power-law relationship, a change in power of one quantity varies as a power of another, resulting in a line in a log-log plot. Power law and log-normal distributions connect naturally, and similar generative models can lead to one or the other distribution depending on minor variations (Mitzenmacher, 2003). The right tail of a log-normal distribution often follows a power law, so, in practice, the distinction between log-normal and power law distributions is often not trivial. If the variance in the left tail is large and “noisy” or thresholded at an arbitrary value, a log-normal distribution may appear as a line in a log-log plot, which is the telltale of the power law. However, a log-normal distribution has a finite mean and variance, in contrast to a power-law distribution in scale-free systems (Barabási, 2002; Barabási and Albert, 1999).

active neurons occupying the right tail of the plot. There is no “golden mean” or representative average neuron in this distribution because the mean is strongly biased by the fast firing minority. A median, which is the middle value in the distribution, is a bit more descriptive, but the shape of the distribution is more informative than any single value. When we replot the data on a logarithmic scale, we see the familiar bell-shaped curve reflecting a normal distribution of log values.

Scientists are interested in the frequency distribution of their observations for at least two reasons. The first reason is pragmatic. The distribution of values tells us how to verify the validity of the observations with appropriate mathematical methods because the choice of these methods depends on the nature of the distribution. There are numerous, so-called parametric methods (such as analysis of variance or regression) to quantify whether two or more normally distributed groups differ.²⁵ The second reason is important for a different goal: a distribution can offer clues about the mechanism that gives rise to those observations.²⁶ If the frequency distribution shows a symmetric bell-shaped curve, we can be confident that the data are produced by a mechanism in which many small independent factors are added to or subtracted from a representative mean value. Likewise, a bell-shaped distribution of log numbers is produced by multiplying or dividing random factors. I discuss the neuronal mechanisms that underlie skewed firing rate distributions later in this chapter. However, first I present a list of structural and dynamic variables in the brain to convince the reader that the log rule prevails in neuronal systems.

25. The frequency distribution of the data is a major factor in determining the type of statistical analysis that can be validly carried out on any data set. Yet frequency distribution is rarely tested rigorously, simply because there are not enough data in most cases. If the variance of a log-normal distribution is small compared to its mean, it can look very similar to a normal distribution. Some statisticians argue that the normal distribution is simply a special case of the log-normal distribution (Lyon, 2004).

26. In 2016, the news broke that the world's largest pumpkin was grown in Belgium. The super squash weighed more than a ton (2,624.6 pounds), easily a hundred times bigger than the ones my family carves for display at Halloween. Distribution of fruit and flower size has long been known to fit a log-normal rather than a Gaussian pattern, and the shape of the distribution has been used to argue that genes combine multiplicatively (Groth, 1914; Sinnott, 1937). Galton (1879) pointed out that variance is the “charms of statistics.” He was the first to introduce the log-normal distribution, although it was not called log-normal until recently. Galton also showed that the response of the eye to light intensity changes follows a log-normal form (confirming the Weber-Fechner law; as described in Koch, 1966).

LOG ARCHITECTURE OF NEURONAL NETWORKS

Brains are among the most sophisticated scalable architectures in nature and come in a variety of sizes from super small to extra large in different species. *Scalability* refers to a property that allows the system to grow yet perform the same desired computation, often with increased efficacy. The largest brain, belonging to the sperm whale (7 kg), is several thousand times larger than that of the smallest mammal, the Etruscan shrew (65 mg). This huge variability raises the question of whether all brains share similar architectural rules. If they do, what are the brain's scaling laws and constraints?

Axon Diameter and Conduction Velocity

One important constraint discussed earlier is the preservation of brain rhythms in mammals (Chapter 6). Thus the computational and syntactic rules, set by these rhythms, should remain the same, irrespective of brain size.²⁷ In other words, neuronal networks need to preserve the brain's ability to temporally integrate large numbers of distributed local processes into globally ordered states so that the results of local computations can be broadcast to widespread brain areas simultaneously. In the reverse direction, local computation and the flow of signals to multiple downstream targets are under the control of global brain activity, often called “executive” or “topdown” control. A critical requirement for effective local–global communication is that the results of local computations in multiple areas are delivered within the integration time window of downstream reader mechanisms set by the various network oscillations. As brains grow larger, their parts inevitably become more distant from each other. To maintain temporal integration across brains of different sizes, hardware solutions are needed to compensate for the longer distances of transmission.

Long-distance communication occurs via the axon, which allows action potentials to travel from neuron to neuron via a synapse. Some axons can conduct spikes more quickly because myelin coats and insulates them, allowing the electrical impulses to move more efficiently. Because axon diameter and myelination determine the conduction velocity of neurons, evolutionary adjustment of these variables appears to be most important for brain size-invariant

27. In scalable systems, certain aspects of the system must be constrained if the same computational goals are to be achieved in the face of increasing organismal complexity (Buzsáki et al., 2013).

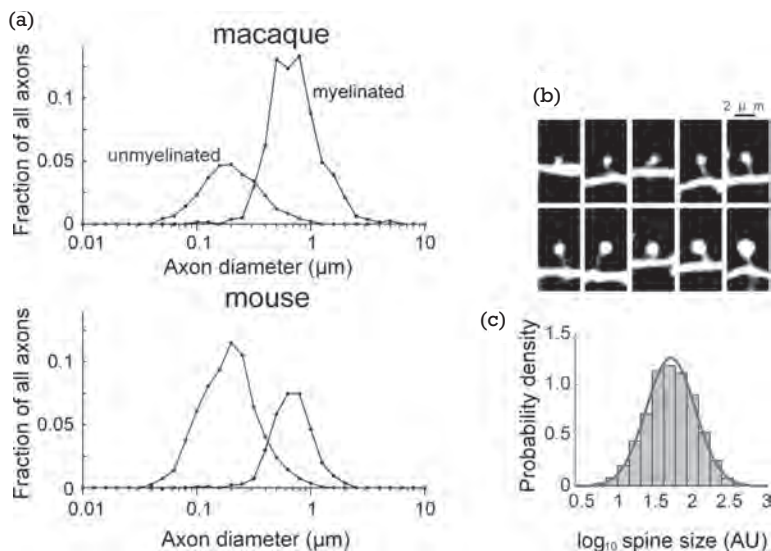


Figure 12.2. A: Distribution of diameters of unmyelinated and myelinated axons in the corpus callosum of the macaque monkey and mouse. B: Examples of spines imaged along the dendrites of one neuron. C: Lognormal distribution of spine sizes. A: Reproduced from Wang et al. (2008); C: Reproduced from Loewenstein et al. (2011).

scaling. Increasing axon diameter enables signals to travel longer distances within a given time window and ensures that signals from various sources can be delivered to the target around the same time.

Axon diameters in the brain vary over several orders of magnitude, and their distribution is strongly skewed. In humans, the great majority of callosal axons have diameters smaller than 0.8 μm , but the thickest 0.1% of axons can have diameters as large as 10 μm (Figure 12.2). Large-diameter axons are typically found in the cross-hemisphere paths that carry sensory information, whereas in frontal cortical areas (where the speed of communication is slower), small diameter axons dominate. The thickness of axons emanating from the same neurons but targeting different brain regions can vary substantially, exemplifying a complex system in which lines of communication have a variety of geometrical and time-computing properties.

If all axons in larger brains increased proportionally in diameter, brain size and energy demands would increase enormously. Instead, a minority of axons with disproportionately increased diameter keeps timing relatively constant across brains of different species. Indeed, the thickest diameter tail of the distribution scales best with brain size. Adding a small fraction of axons with very large diameters can preserve cross-hemisphere conduction times across species

(Figure 12.2).²⁸ In summary, a disproportionate increase in larger diameter axons with fast conduction velocities keeps communication speed similar as brain size increases.

Microscopic Connectivity

The efficacy of communication between neurons, called *synaptic transmission strength*, depends on the anatomical features of the synapse. Excitatory inputs onto other excitatory neurons form synaptic connections with microscopic protrusions of the dendrites, known as *spines* because of their appearance under the microscope. Drumsticks or mushrooms would be perhaps more appropriate, as the head of the spine is connected to the dendrite by a narrow neck. This head is contacted by the presynaptic bouton at the end of the incoming axon. The volume of the spine is an important measure because it is correlated with the amount of postsynaptic density, which in turn is reliably correlated with the amplitude of the excitatory postsynaptic currents induced by the presynaptic release of excitatory transmitter.

Microscopic imaging studies have documented a large variety of spine sizes on single pyramidal neurons. Giant spines can be several hundred times larger than the smallest ones. The distribution of spine sizes in individual layer 5 neocortical neurons and hippocampal pyramidal cells is well described by a log-normal rule²⁹ (Figure 12.2). This large variation implies that some partner neurons exert a strong influence on the receiving neurons via large synapses, whereas most upstream partners establish weak synapses and therefore do not have much influence individually. This situation is thus quite similar to our social connections. A few individuals have a very strong impact on our behavior, while most of our acquaintances have little or no influence.

Mesoscopic Connectivity: A General Wiring Rule

If the brain were a *tabula rasa*, neurons randomly connected to each other would do a good job. Most network models use random connectivity

28. Swadlow (2000); Aboitiz et al. (2003); Wang et al. (2008); Innocenti et al. (2014). The host neurons of the giant caliber axons still need to be identified. At least a fraction of them are likely to be inhibitory neurons; the myelinated axon diameter of long-range inhibitory neurons in the rat can reach 3 μm (Jinno et al., 2007), several times thicker than that of pyramidal cells. In turn, theoretical and modeling studies suggest that long-range interneurons are critical for brain-wide synchronization of gamma, and potentially other, oscillations (Buzsáki et al., 2004).

29. Yasumatsu et al. (2008); Loewenstein et al. (2011).

statistics. Experience then could modify the connection strengths, as is often the case in artificial neural networks where every neuron has the same probability of connecting to every other neuron. In contrast to this expectation, careful microcircuit analyses in several laboratories have repeatedly demonstrated that neuronal motifs, such as that two neurons are connected reciprocally or that interconnected neurons often share inputs from the same third neuron, occur more frequently than expected from a random graph map.³⁰ However, general rules by which some such nonrandom connections are formed and how such rules scale across brains of different sizes have not been fully addressed yet.

Anatomical studies in macaque monkeys that trace the axonal projections back to their cell bodies show that the range of connection strengths between brain areas spans five orders of magnitude. A given cortical area is connected strongly to a few structures and weakly to many other structures. This connectivity profile is best described by a log-normal distribution.³¹ Brain anatomy is similar in the mouse brain. An industrial-scale effort at the Allen Institute in Seattle showed that half of all axons sent out from each cortical area are directed toward only a handful of other areas. These few connections may represent half of all axons sent out from the cortical area in question. The remaining half of the axons weakly innervate a large number of areas. As in the monkey, the distribution of connections follows a log-normal form. In fact, the log rule applies not only to intracortical connections but to all the connections of any cortical patch, including their thalamic and other subcortical targets. Although this pattern has been established quantitatively in only two species so far, it is tempting to suggest that scaling of interareal connections is conserved across species and that the strength of interactions between regions obeys similar mesoscopic wiring rules.³² With this information in hand, we can speculate how these structural foundations affect spike-mediated communication across neuronal assemblies.

30. Song et al. (2005); Yoshimura et al. (2005); Perin et al. (2011); Ko et al. (2013).

31. Markov et al. (2011, 2013); Wang et al. (2012); Ercsey-Ravasz et al. (2013). These findings considerably modify previous connectivity estimates in the neocortex. The average path length had been considered between 3 and 4, whereas the log-normal connectivity suggests that it is approximately 1.5. In other words, any cortical area is connected with any other through just one or two synaptic relays, although the long-distance excitatory connections are typically very weak.

32. Oh et al. (2014).

LOG DYNAMICS

As expected from the log-normal distribution of spine volumes in pyramidal cells, there is also a large variation in synaptic weights, which reflects the strengths of physiological connections among neurons. The most accurate way to assay connection strength is to record from pairs of neurons intracellularly and quantify the amplitude of the excitatory or inhibitory postsynaptic potentials in the postsynaptic neuron in response to evoked spikes in the presynaptic neuron. Because multiple intracellular recording is not practical in behaving animals, indirect methods are used to estimate synaptic strength. These involve calculating the probability of spike transmission between synaptically connected pairs of neurons (Figure 12.3). Typically, these two types of methods show good agreement.³³ Such experiments have established that spike transmission probability between neurons shows a log-normal distribution, and spike transmission, in general, is stronger in waking than in sleeping animals. Stronger synapses are not only stronger on average but also more reliable from event to event, reminiscent of the Weber-Fechner logarithmic law.

Log-normal Distribution of Firing Rates and Spike Bursts

The firing rate distribution of hippocampal neurons in Figure 12.1 is representative of all known neuron types in all cortical and subcortical regions of every species investigated to date, from turtles to humans.³⁴ The mean spontaneous firing rates of individual neurons of the same class span four orders of magnitude. In any given time window, 10–20% of neurons within each type contribute half of all spikes, whereas the rest are responsible for the remaining half of the spikes. These distributions can shift to the left or right across different classes of neurons and in different brain states, matched by similar shifts in several types of inhibitory interneurons. Importantly, the log firing rates of spontaneous spikes and spikes evoked by stimuli strongly correlate. If a stimulus evokes an extra spike in a neuron firing at a rate of one per second, the

33. Gerstein and Perkel (1969); Csicsvari et al. (1998); Fujisawa et al. (2008). To verify synaptic connections between neuron pairs, English, McKenzie et al. (2017) decoupled presynaptic neurons from ongoing network activity through single-cell juxtacellular current injection or optogenetic stimulation of small groups of cells in behaving mice. For direct paired recordings *in vitro*, see Song et al. (2005). Inhibitory neurons can also serve as hubs by either integrating larger numbers of inputs or contacting many targets in wide areas (Bonifazi et al., 2009).

34. Sometimes, rate distributions show a better fit with gamma than with log-normal distribution.

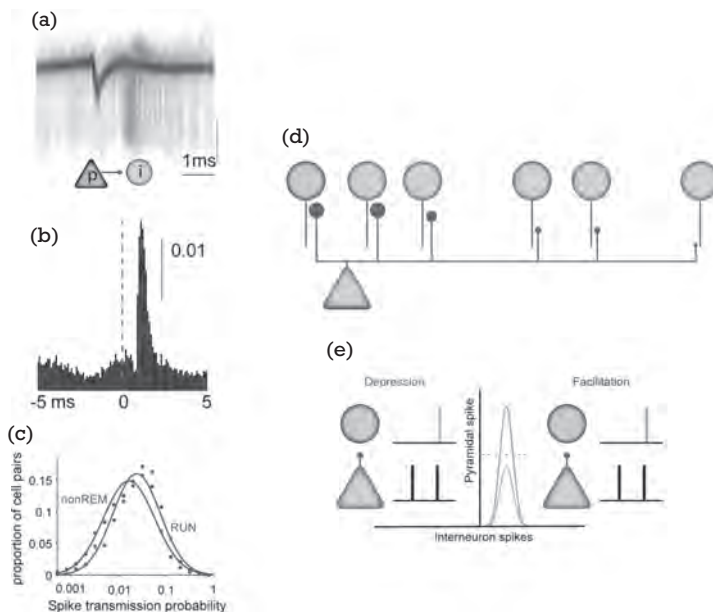


Figure 12.3. Dynamics of pyramidal cell–interneuron synaptic connections.

A: Superimposed spikes of a pyramidal cell (*p*, small wide spikes) and an interneuron (*i*, large narrow spikes) triggered by spiking of the pyramidal cell. Note many spikes of the interneuron at 1 msec after the spike of the pyramidal cell. B: Temporal offset of spike transmission probability values between the CA1 pyramidal cell spikes (time 0) and spikes of the interneuron (peak of the histogram). C: Distribution of spike transmission probabilities (note log scale) during exploration (RUN) and non-rapid eye movement (non-REM) sleep. The right shift of the curve indicates stronger spike transmission during exploration. D: Connection strength decreases rapidly with distance between the neuron pairs. E: Spike transmission probability exhibits both activity-dependent short-term facilitation and depression, as indicated by the Gaussian curves above and below ‘no change’ (horizontal dashed line).

Modified from English, McKenzie et al. (2017).

same stimulus may induce ten excess spikes in another neuron whose baseline firing rate is ten per second. This is a huge difference in the number of induced spikes, yet proportionally they yield the same value. In other words, the response of a neuron to extrinsic inputs is proportional to its long-term firing rate, reminding us of the Weber-Fechner law.³⁵

35. Remarkably, fast firing neurons also respond more robustly to optogenetic stimulation when the only or at least main source of excitation is channel rhodopsin-controlled depolarization. The firing rates of the artificially (light) induced spikes and spontaneous firing rates strongly correlate (Stark et al., 2015).

When the intervals between spikes are short, say 10 ms, that firing event is called a spike *burst*. The fraction of these short intervals relative to all interspike intervals is a quantitative measure of a neuron's burst propensity. Burst distribution also has a log-normal form. A small fraction of pyramidal neurons at the right tail of the distribution can be called *superbursters*, emitting a burst 40 to 50% of the time, whereas those at the left side hardly ever burst.

Skewed Distribution of Ensemble Size

The impact of a neuronal assembly on its targets depends on its degree of synchrony. There is a difference between a hundred neurons firing together in seconds or in a short gamma cycle (Chapter 4). Temporal synchrony can be measured by quantifying the fraction of spiking neurons in any given time window. Such measurements show that “ensemble size”—the number of neurons active in a particular time window—does not vary around a typical mean. Instead, the magnitude of synchrony follows a log-normal distribution: rare strongly synchronized (very large) events are interspersed irregularly among many medium and small events.³⁶ The skewed nature of the distribution implies that there is no characteristic neuronal ensemble size in cortical networks.³⁷ Rapidly firing neurons are more frequent participants in large population events as more frequently occurring spikes are expected to coincide more often with any other event than spikes of slow neurons.³⁸ However, what

36. For example, a small minority (1.5%) of CA1 pyramidal cells participates in half of hippocampal sharp wave ripple events, whereas half of all neurons fire in less than 10% of the sharp wave ripples (Mizuseki and Buzsáki, 2013).

37. The log-normal distribution of population synchrony is a challenge to Hebb's definition of cell assemblies. If the postulated assembly is a fixed set, it is not clear why the fraction of spiking neurons in successive time windows varies by orders of magnitude. However, if a good part of the neuronal messages can be carried by a minority of highly active neurons, then in each time window “good enough” information is present. Not only the fraction of active neurons but the correlation coefficients between neuron pairs also shows log-normal distribution, and coactivated neuron pairs largely correlate with the firing rates of the partners (Buzsáki and Mizuseki, 2014). Coactivation of calcium signals of CA3 neuron pairs in hippocampal tissue culture also shows a log-normal distribution (Takahashi et al., 2010). The factors that give rise to the strongly skewed distribution of population responses are not well understood. The log-normal variation of synaptic weights is a likely source because computational networks with dynamic synapses can respond robustly to external inputs yet return to baseline activity shortly after the perturbation (Sussillo and Abbott, 2009).

38. Okun et al. (2015) found that neighboring cortical neurons can differ in their coupling to the overall firing of the population. They called strongly coupled ones “choristers” and weakly coupled ones “soloists.” It is possible that choristers and soloists represent neurons drawn from

is even more critical is their better connectedness to each other and everyone else in the population. This oligarchic “hub” nature of connectedness makes fast firing neurons more frequent partners and often leaders of large population events. Not only the magnitude but the duration of population events is also log-normally distributed, including that of hippocampal sharp wave ripples, neocortical slow oscillations, and thalamocortical sleep spindles.

Autocratic Organization of Neural Networks

The skewed distributions of so many structural and dynamic variables in neuronal networks constrain their operations in important ways. The fast firing minority has a special status. They not only work harder all the time, but they also have many more connections as well as being more strongly connected to their target partners, including other fast firing neurons.³⁹ This is the case for both excitatory and inhibitory neurons. They form *hubs*—an oligarchic organization of connections and information transmission. In network speak, such partnership is called a “rich club.”⁴⁰ The highly active autocratic neurons have longer axons and send their axons to more structures than their slowly firing peers. Through these connections, the privileged club members have access to more information than nonmember neurons, and they share such information among themselves.

Again, it is important to reiterate that, despite the hard work of the hub neurons, they contribute only, say, half of the spikes and exert half of the influence. However, for normal brain operations, the other half is equally important. The remaining spiking activity is supplied by a large fraction of slowly discharging neurons, which are connected via weaker synapses in a more loosely formed large network. Although we like to anthropomorphize and talk about hubs and leaders, it is critical to emphasize that in the log-normal oligarchy, there is no definable demarcation line between the fast and slow and strong and weak ends. Therefore, the boundary between local and global connectivity

the right and left tails of the log-normal rate distribution, respectively. Population coupling is largely independent of sensory preferences, and it is a fixed cellular attribute, invariant to stimulus conditions. As expected from features of high-firing neurons, choristers respond to multiple features and are connected with larger numbers of other neurons compared to soloists (Mizuseki and Buzsáki, 2014).

39. Yassin et al. (2010); Ciochi et al. (2015).

40. Olaf Sporns’s book on brain networks (2010) has many links relevant to my discussion of the skewed organization of neuronal networks. See also van den Heuvel and Sporns (2011); Nigam et al. (2016).

is opaque. The advantage of networks with such local–global interactions is an optimal balance between computation speed and accuracy.

As it should be clear from the preceding discussion, a measurement as simple as the long-term firing rate goes a long way because rate is related to so many other features of the neuron. A pertinent analogy here is language. The frequency with which different words are used in everyday language affects their rate of meaning change across all languages and time scales. There is almost a hundredfold variation in rates of such lexical evolution among words in Indo-European languages, and the distribution shows a log-normal-like form. The most frequently used words, such as “one,” “two,” “who,” and “night” are strongly conserved so it may take 10,000 years to diverge across languages, whereas rarely used words, such as “to turn,” “to stab,” and “guts” may change in just a few hundred years.⁴¹ The frequently used words, similar to fast firing neurons, form rigid hubs and dominate our everyday conversations, while many words of our existing vocabulary are rarely used. It is tempting to conclude that the brain mechanisms that support such selection are related to the log rules that operate in brain structure and dynamics. How are such distributions generated and maintained, and what are their functional consequences?

CREATING LOG DYNAMICS

Perhaps the most exciting question is how the log patterns at so many levels of brain organization relate to each other. Skewed distributions can arise by a variety of nonlinear operations. Thus it is not obvious how the log distribution of anatomical connectivity gives rise to log dynamics or how skewing of synaptic weights relates to log-normal firing rates and skewed population cooperation. There are at least two options. Even if the inputs to a population of neurons show a bell-shaped distribution, the firing rate distribution of the responding neurons can become skewed due to nonlinearities in their membrane properties. Alternatively, the neurons may respond linearly to a log-normally distributed synaptic input, in which case the source of the skewness is the input.⁴²

41. I borrowed these examples from Pagel et al. (2007), which describes several further consequences of rate of occurrence. The rate of word usage may be analogous to the externalization of brain processes. The more frequently we encounter similar consequences of an action, the more explicit its brain representation becomes.

42. Several papers have dealt with this topic: Hromádka et al. (2008); Koulakov et al. (2009); Roxin et al. (2011); Ikegaya et al. (2013).

As discussed earlier, the strengths of the hundreds to thousands of synapses on a single pyramidal cell have a log-normal distribution. Yet, given that a pre-synaptic neuron alone rarely causes a spike in a postsynaptic cell, we also need to know the rules of presynaptic cooperation and postsynaptic integration of inputs. We also learned that the magnitude of population synchrony follows a log-normal pattern. Yet even this combined knowledge is not sufficient to predict how neurons will respond without knowing their integration rules. Indirect evidence and modeling results support the first hypothesis; namely that the log-normal distribution of firing rates across neurons emerges from the skewed distribution of their intrinsic, nonlinear properties. While it is clear that variation in firing rates is coupled to variation in synaptic efficacies, stronger evidence is needed to establish which one is of primary importance and drives the second.⁴³

We can only speculate how skewed distributions emerge in the first place. During neurogenesis, early-born neurons have a higher chance of innervating each other, and the synapses they make on the budding dendrites are placed nearest to the soma. Thus later-born neurons can only innervate the more distant segments of the dendrites. Based on this observation, we can conjecture that early-born neurons are more effective in discharging their peers, so the firing rate distribution may have a simple chronological origin.

Maintaining Log-Dynamics

How are these firing patterns maintained? Consider the following two options and guess. First, after the fast-firing neurons get tired, they take a long nap while previously dormant ones are invigorated, keeping the overall activity distribution the same but with revolving active participants. Second, the rate distributions remain correlated across brain states and across various testing situations, implying that some neurons must work harder than others forever. If you cannot decide, no worries. When I asked many of my colleagues, the majority favored the first option, but some argued in favor of the second, which is correct. Individual neurons maintain their firing rate ranks over days, weeks, and months, as if they sense their own firing outputs and adjust them to a set point customized to each cell. The distribution of intrinsic firing rates reflects a fundamental biophysical heterogeneity in neuronal populations, of which

43. In a neural network model, it is the spiking rate that seems to be of primary importance (Kleberg and Triesch, 2018). When spike timing-dependent plasticity rules and multiplicative normalization (Chapter 11) are incorporated into the network, the initially similar synaptic weights will soon form a skewed distribution and hub neurons with strong outgoing synapses.

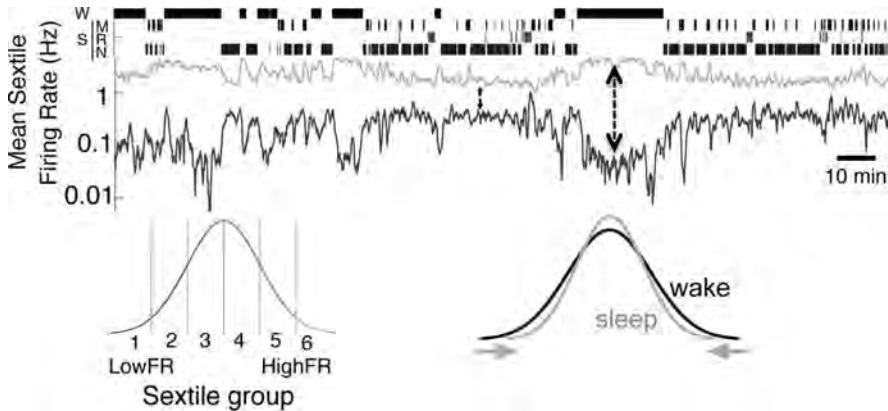


Figure 12.4. Sleep and wake states oppositely affect fast and slow firing neurons. *Bars:* Wake–sleep state scoring. W, wake; S, sleep; R, rapid eye movement (REM) sleep; N, non-REM sleep; M: microarousal. Firing rate of fastest and slowest sextile groups over wake and sleep change in opposite directions (*large and small double-headed arrows*; sextile distribution is shown in the left inset). *Bottom right:* Log-gaussian curves illustrate the log firing rate distributions at the end of waking period (*black*) and at the end of sleep (*gray*). Sleep narrows the rate distribution of firing rates.

the neuron's firing rate acts as a readily identifiable marker.⁴⁴ Of course, the firing rates of individual neurons vary transiently (e.g., in response to relevant stimuli), but the magnitude of the response is correlated with the neuron's long-term default rate. Cortical pyramidal cells can increase their spiking activity up to about 80/s. However, when the rate of a neuron is sampled over a long period of time, it will be dominated by the activity in its “idling” state.

The neurons' firing rate rank in the population is maintained through sleep. Neuronal excitability, as expressed by firing rates, and local network-related sources of neuronal heterogeneity, as expressed by the skewed distribution of synaptic weights, are related. This relationship arises from, and is maintained by, neuronal dynamics during alternating sleep–wake cycles (Figure 12.4).

44. Even after a sustained change in the firing rate due to altered afferent drive, the neuron returns to its previous rate (Koulakov et al., 2009; Mitra et al., 2012; Mizuseki and Buzsáki, 2013; Fu et al., 2016; Hengen et al., 2016). The log-normal distribution of spike burst rates favors the idea that membrane channel properties are important because the likelihood of burst occurrence depends largely on the intrinsic properties of neurons. Yet, in addition to neuron-autonomous rate homeostasis regulating intrinsic biophysical properties and channel density variations, the networks in which neurons are embedded can also exert a sustained influence on neuronal spiking.

Sleep has long been viewed as a brain state necessary to recuperate exhausted neurons from their hard work during waking.⁴⁵ The details of this hypothesized “sleep homeostatic plasticity” are still debated, but inroads have been made. The regulatory forces of sleep serve to restore population-wide dynamics to maintain a stable neuronal system. During sleep, release of subcortical neuromodulators is diminished, the neocortex is isolated from the influences of the outside world, and the thalamocortical system sustains its activity in a self-organized manner.⁴⁶ Sustained firing of a neuronal population (called an “up state”) is interrupted by brief (30–200 ms) population-wide silent periods (“down states”). During the down state, nearly the entire cortex, thalamus, and many subcortical sites are silent. The brain has to reboot itself from this silence thousands of times each night. The durations of both the up and down states follow a log-normal distribution, and their relative durations shift over the course of a night’s sleep.⁴⁷

Experimental observations show that, during sleep, neuronal population activity transitions from an externally driven to an internally driven regime, complemented by a global increase in drive from the local network. During waking, the log-normal distribution of firing rates widens, meaning that many neurons at the left tail become silent or reduce their discharge activity, while the fast-firing neurons increase their spiking. During non-rapid eye movement (non-REM) sleep (i.e., the greater part of sleep when saccadic or rapid eye movements do not occur), the opposite happens. Many wake-silent neurons become active, while the very active neurons gradually decrease their firing rates over the course of sleep. As a result, the brain wakes up with a much narrower and sharper log-normal curve than before sleep.

What mechanisms could widen and narrow the firing rate distribution? As discussed in Chapter 11, subcortical neurotransmitters modulate gain and control the responsiveness of cortical neurons to sensory inputs. Thus, sensory

45. In contrast to memory consolidation (Chapter 8), in which specific synapses are thought to be “selectively” modified to strengthen particular memory traces, homeostatic function is assumed to involve a general, population-wide modification of synapses to maintain a stable neuronal system. An influential model by Tononi and Cierelli (2003, 2014; Vyazovskiy and Harris, 2013) suggests that most active cells and synapses during waking continue to be active during sleep, whereas weak synapses and slow-firing neurons are “down-selected” (i.e., eliminated from network participation). The model would predict, therefore, that log-normal distribution would be asymmetrically affected during sleep by reducing, rather than increasing, the rate of slowly firing neurons (i.e., opposite to the experimentally observed changes).

46. Pioneers of sleep research suggested that the power delta activity during non-REM sleep is a reliable measure of sleep pressure and serves as a homeostatic mechanism (Feinberg, 1974; Borbely, 1982).

47. Watson et al. (2016); Levenstein et al. (2017).

drive could explain the elevation of firing rates among fast neurons during waking. The faster responding neurons more effectively recruit inhibitory neurons, which in turn would further reduce the firing rates of the already slow neurons. As a result, both ends of the distribution would widen.

Another way to widen the rate distribution is via synaptic plasticity. At least one mechanism, called *spike timing-dependent plasticity*, depends on the temporal order of spikes. If neurons *A* and *B* are mutually connected, and *A* consistently fires before neuron *B* within the duration of a gamma cycle, the synapse *A* to *B* will be strengthened but *B* to *A* will be weakened (Chapter 4). Because the probability of temporal coincidence increases with the firing rates of the coactive neurons, this synaptic plasticity rule favors preferential strengthening of synapses involving neurons with higher firing rates. In the waking brain, the asynchronously active heterogeneous population leads to asymmetric strengthening of synapses on neurons with higher spontaneous firing rates. The consequence of this process is easy to imagine: excitable neurons become progressively more excitable, which can eventually destabilize the network. Unless some mechanism combats this “plasticity pressure” in the awake brain, the resulting positive feedback loop could increase synaptic weights to maximum saturation. One mechanism that counteracts this pressure is the up-state transition of non-REM sleep.

During the transitions from down state to up state in non-REM sleep, neurons fire in a sequence. A neuron’s place in this order is correlated with its baseline firing rate, such that neurons with higher firing rates tend to spike before those with lower firing rates (Figure 12.5).⁴⁸ A simple but important consequence is that neurons with high and low firing rates get temporally segregated. Because high-firing neurons tend to fire earlier than low-firing neurons after the down-to-up transition, the plasticity rule tends to increase the weights of synapses from high-firing onto low-firing neurons while decreasing the weights of synapses from low-firing onto high-firing neurons. This redistribution of synaptic weights during non-REM sleep pulls both ends of the log-normal firing rate distribution closer to the mean and acts as a homeostatic counter to changes in the asynchronous wake state.⁴⁹ This normalization mechanism is an important function of sleep.

48. Luczak et al. (2007); Peyrache et al. (2011); Watson et al. (2016); Levenstein et al. (2017).

49. The skewed organization of neuronal circuits has important implications for disease. Homeostatic maintenance of the status quo between the two ends of the distribution is critical for the normal brain. A shift to either direction will inevitably impair performance. For example, stronger than needed dominance of the hub neurons can form an epileptic focus.

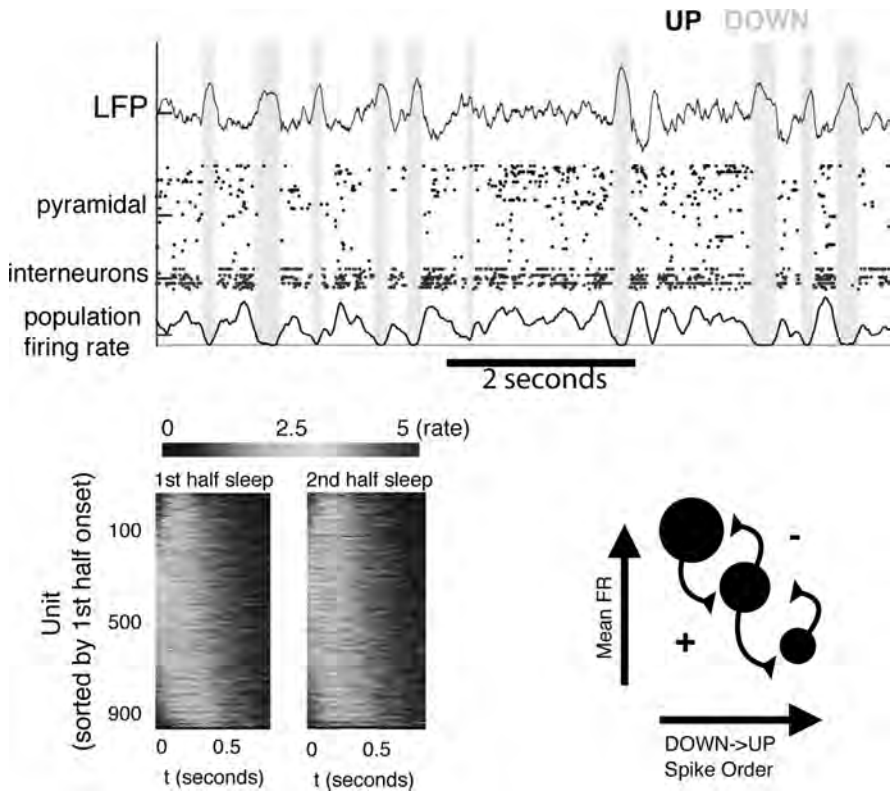


Figure 12.5. Homeostatic regulation of firing rates by “down” states of non-rapid eye movement (non-REM) sleep. *Top:* Illustration of up and down states. Note that during down states both pyramidal cells and interneurons are silent (dots spikes of neurons; each row is a single neuron). *Bottom left:* Sequential activity from high to low firing rate cells at the down–up transitions of non-REM sleep (> thousand neurons). Each gray line is the firing event of a single cortical pyramidal neuron, sorted by the occurrence of its first spike after the up state onset, shown separately for first and second halves of non-REM sleep. The density of gray reflects the firing rate of the neuron. Note that faster firing neurons tend to fire earlier at the up state onset (neurons on the top). *Right:* Spontaneous plastic pressure between low (*smaller circles*) and high (*larger circles*) firing rate cells during non-REM sleep due to high-to-low firing rate sequential activity at the down–up transition. Synaptic plasticity rules increase and decrease the firing rates of slow and fast firing neurons, respectively.

From Watson et al. (2016); Levenstein et al. (2017).

CONSEQUENCES OF LOG RULES

I began this chapter by explaining that skewed connections and activity dynamics are necessary to maintain the status quo among competing properties of neuronal networks.⁵⁰ Next, I will illustrate that a wide range of firing rates, synaptic weights, and degrees of population synchrony provides several advantages.

Familiar–Novel Continuum

The terms “old and new” or “familiar and novel” are designated as discrete categories, yet they may also be viewed as the two extremes of a continuous quantity. Log-normal distributions may explain how these categories are represented in neuronal networks because the two ends of the distributions are associated with multiple distinguishable properties.

In novel and familiar environments, most hippocampal pyramidal neurons have no or only one place field, while a small minority of neurons have many. Thus, the distribution of maze coverage is strongly skewed. When a hippocampal neuron does not fire in a given maze, we expect that it may fire in another one. That is, every hippocampal neuron is viewed as a potential place cell. This idea was tested by training rats to run novel maze tracks with lengths of 3, 10, 22, and 48 meters. Some place cells that fired on the short tracks formed additional fields on the larger tracks, but most new place cells were recruited from the pool of initially silent cells. The number of fields formed was strongly skewed: a few neurons had many fields, whereas many neurons had one or none, similar to the behavior of hippocampal neurons in a single maze. Extrapolation from the observed log distributions suggested that nearly all hippocampal pyramidal cells would be active in an environment with a diameter of approximately 1 kilometer, which is believed to correspond to the ecological niche of rats. In another experiment, rats were tested in multiple rooms. Most pyramidal neurons fired only in a single room, but a small minority fired in multiple rooms or all rooms, producing

50. The *spike timing-dependent plasticity* (STDP) rule can segregate synaptic weights into a bimodal distribution of weak and strong synapses, but the network may destabilize without additional constraints. However, when STDP works in a synaptic weight-dependent manner (called log-STDP), it can produce a skewed weight distribution and sustain stable network dynamics. In such a regime, long-term depression exhibits linear weight dependence for weak synapses but much less so for strong synapses. The long-term depression of strong synapses enables synaptic weights to grow, generating a long tail in the synaptic weight distribution (Omura et al., 2015).

a log-normal distribution of the overlap of neuronal activity in the different rooms.⁵¹ Overall, these experiments demonstrate that the skewed distribution of place fields is a general rule, irrespective of the nature or size of the testing environment.

The generalizer–specialist continuum is not an exclusive feature of higher order cortical areas but is also prominently present in primary sensory cortices. The majority of neurons in the superficial layers of the primary visual cortex respond relatively selectively to orientation of the stimulus (e.g., vertical or horizontal bars) and/or to the direction of the moving stimulus (e.g., left to right or right to left). Some neurons are true specialists and respond selectively to a single feature. However, across the population, the magnitude of selectivity varies considerably. A small group at the right end of the distribution responds to several visual features. Similarly, a minority of neurons in the somatosensory cortex shows selective preference to specific directions of whisker deflection. Yet another minority responds to multiple directions, with most neurons responding with intermediate specificity. In the auditory cortex, superficial neurons can have highly specific and narrow frequency tuning properties (such as one neuron selectively responding to a 10 kHz beep, whereas another responds to a 20 kHz beep), where others can respond to multiple overlapping bands. The overall wisdom that we can draw from the many physiological experiments performed in numerous laboratories and in different parts of the cortex is that some neurons may appear to treat multiple stimuli and situations as the same and similar; that, is they generalize input features. Other neurons, on the other hand, appear to be super-specialists and respond only to a single feature of the many options. However, the most important message of this discussion is that the generalizers and specialist form a continuum.⁵² Whether an object or situation is judged as the same, similar, or different depends on how the downstream observer neurons and circuits sort out their responses to the spike output of a skewedly distributed upstream population.

Nonegalitarian Votes of Neurons

What could be the advantage of such skewed distributions, for example, for ‘encoding’ spatial relationships? From the perspective of independent (also called “orthogonal” in geometry) coding, the minority of neurons with multiple

51. Alme et al. (2014); Rich et al. (2014). For a review, see Buzsáki and Mizuseki (2014).

52. Ohki et al. (2005); Rothschild et al. (2010); Kremer et al. (2011).

place fields may be regarded as “noise” or imperfection in the system.⁵³ However, when other physiological features of this minority are considered, a different picture emerges. Not only are they more active in multiple environments, but their firing rates also are higher, they emit more spike bursts, and their individual place fields are larger compared with those of the majority neurons. The higher mean firing rates of the active minority in the maze strongly correlate with their peak firing rates within their place fields and even with their firing rates during sleep in the animal’s home cage. Furthermore, the diligent minority fires synchronously with other neurons more frequently across all brain states than the slower firing majority and produces a stronger and more effective excitation in their targets. As a result, in physiological time frames, such as theta oscillations and sharp wave ripples (Chapters 6 and 8), about half of the spikes in the hippocampus are contributed by this active minority, with the other half from the majority of neurons with no or a single place field.

Thus, the neurons in cell assemblies make strongly skewed contributions, suggesting that the cell assembly is a nonegalitarian organization. For example, the consensus among the low-firing majority of hippocampal place cells could be that “the owner of the brain is now in a different room.” If votes are represented by spikes, each of them may have only a single vote, although there are many voters. In contrast, the high-firing minority may represent the view that “this is the same room.” Despite their small number, they have many spikes, possibly stronger synapses, and thus many votes. Downstream reader neurons have to interpret this mixed output to make a judgment and generate action potentials (Figure 12.6). The high-firing neurons may exert a stronger influence by emitting more spikes within the same time window. But we should also consider the option that the reader neuron responds to the ratio of change; that is, it normalizes the spike messages from both fast and slow neurons, in which case the proportional impact of the two upstream neuronal populations is equal.

Rigid-to-Plastic Continuum

If neurons with different activity rates exert a nonegalitarian influence on their downstream partners, new experience might also have a differential effect on the two tails of the log-normal distributions. In addition to homeostasis, non-REM sleep is also critical for memory consolidation. As discussed

53. The idea of orthogonalization was introduced to neuroscience by Marr (1969) to support his hypotheses that independent constellations of neurons can represent the numerous stimuli we encounter.

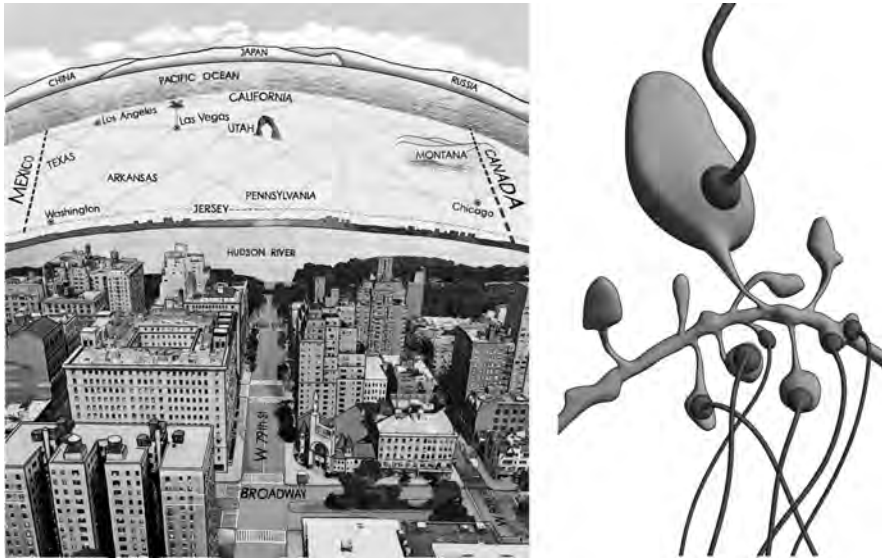


Figure 12.6. *Left:* New Yorkers’ skewed view of the world, inspired by Saul Steinberg’s 1976 illustration in the *New Yorker* magazine. *Right:* Neurons receive mixed messages from upstream partner neurons. The majority of weakly active neurons convey detailed information about the distinctiveness of a situation, whereas the strongly firing minority with strong and rigid synapses generalize across situations.

in Chapter 8, memory replay is often conceptualized as occurring via higher order interactions within populations of neurons that have similar properties. However, as demonstrated earlier, cortical networks contain neurons with highly diverse biophysical properties, characterized by skewed distributions of synaptic weights, long-term firing rates, and spike bursts. Furthermore, their temporal correlations are largely preserved across brain states and environmental situations, suggesting that learning-induced changes are constrained within a dynamically stable network.

In the widely used population analysis methods, the individual differences among neurons are obscured. This is because a small group of highly active neurons can strongly bias the conclusions reached by such methods, as is typical of populations with skewed distribution.⁵⁴ To counter this problem, every neuron in the population should be characterized individually. Just as in sport, where one would like to know how a team would perform if one of the players is pulled out, in the team performance of neuronal actions we need

54. This is similar to human interactions: “Remove one highly assertive member from a group of eight candidates and everyone else’s personalities will appear to change.” (Kahneman 2011).

to measure the impact of individual neurons because they are not equal. Such individual neuron analyses demonstrate that learning-induced plasticity is not equally distributed among hippocampal pyramidal neurons after an encounter with a novel spatial task. Instead, the sequences of place cells in the novel environment are formed from neurons that already had place cell features before the experience and do not change much during learning. We can call them “rigid” cells. These rigid neurons belong to the high firing end of the population, and they are more strongly connected to each other. Many of their afferent connections may already be too strong, also called “saturated,” and therefore cannot get much stronger.⁵⁵ At the other end of the spectrum are the “plastic” neurons. They also may have a place field from early on in a new environment, but they can modify their firing patterns and can become incorporated into the backbone of the rigid group after experience.⁵⁶ Thus, the hippocampal network contains a continuum of neurons spanning a range of rigid to plastic features. The plastic cells show lower mean firing rates, more specific firing fields, and larger changes in firing rates and field specificity during the first minutes of exploration than the rigid cells. Fast-firing rigid cells provide a backbone into which slow-firing plastic cells can be rapidly incorporated (Chapter 8).⁵⁷

55. Perin et al. (2011), investigating interactions among neocortical neurons in vitro, have found that synaptic strength correlates with connection probability among neurons. Furthermore, the synaptic strength can saturate after only a small fraction of the maximal clustering is reached. Thus, they comprise a rigid cluster.

56. Groszmark and Buzsáki (2016). The rigid versus plastic ends of the distribution relate to the long debate about whether place fields preexist or require experience (Chapter 13). Early studies suggested that hippocampal pyramidal cells display place fields when the rat first enters a novel environment (Hill, 1978; Samsonovich and McNaughton, 1997). Before our work, Dragoi and Tonegawa (2011, 2013a, 2013b) reported that place cell sequences in a novel part of a maze were correlated with sharp wave ripple-related sequences during sleep before the novel experience. They called such sequences “preplay” (Dragoi and Tonegawa, 2011). The existence of preplay was challenged on statistical grounds (Silva et al., 2015) and by suggesting that the rat’s visual survey of the novel parts of the maze before the animal entered those corridors induced place cell sequences (Ólafsdóttir et al., 2015). This line of reasoning is also supported by other studies showing that stable firing fields require at least a few passes through novel corridors (Wilson and McNaughton, 1993; Frank et al., 2004) and perhaps postexploration sharp wave ripples. In our studies, the rats were tested in an entirely new room. Yet it is possible that visual inspection of the environment before the first run contributes to the selection of preexisting rigid generalizer cells to form a scaffold of sequential activity that unfolds during exploration.

57. Experiments by George Dragoi in my laboratory (Dragoi et al., 2003) demonstrated that place cells with high within-field firing rates are robust against perturbation of the synaptic strengths of hippocampal networks. In contrast, slow-firing place cells had a high propensity to express new place fields after we artificially affected their synaptic inputs (see also Lee et al., 2012; Bittner et al., 2015).

Plastic neurons also have higher place-specific indices compared with rigid cells and typically only one place field. The firing patterns of the plastic and rigid neurons change differently during learning. While both fast- and slow-firing neurons can have place fields from the beginning of maze exploration, slow- but not fast-firing neurons increase their spatial specificity steadily during learning, as measured by the ratio of the number of spikes emitted inside versus outside the place field. This change in behavior is what makes these cells “plastic.” Learning thus does not necessarily create new place fields but increases the “signal-to-noise” ratio, so that the hippocampal spatial map becomes more reliable. Plastic hippocampal neurons also increase their burst emission during sleep after experience, especially when they fire together with other neurons during sharp wave ripples (Chapter 8).⁵⁸ Overall, these experimental findings reveal that replay sequence-forming neurons during both waking and sleep are drawn from a skewed distribution with varying coding and plasticity properties and that replay events send a synthesis of preexisting and recently learned information to downstream observer neurons.

SKEWED DISTRIBUTION OF TERRITORY

So far, the picture I have painted is that skewed distributions of both structural components and dynamics of neurons and neuronal networks may contribute to many important functions of the brain, including perception and memory.⁵⁹ The brain’s skewed organization could also influence our behavior and social interactions. Building a mechanistic link between structural, functional levels of brain organization and behavior, especially social behavior, is acknowledged to be difficult problem. Yet we can look at some examples. Territorial behavior is well studied by ethologists, but quantitative experiments involving large populations are rare. Figure 12.7 illustrates an early attempt to do just that by examining a closed mouse village in which food and water were supplied by the researchers.

Of the fourteen available shelter boxes, one male owned three, another male owned two, and five males owned one each. These males were heavier and stronger than the remaining 21 males. Five of the strong males had one female companion each, whereas two had two companions each. Another eight males lived together in one small box without female partners, and the remaining

58. Preventing place cells from firing during sharp wave ripples interferes with place cell stability (Roux et al., 2017).

59. Fusi et al. (2007).

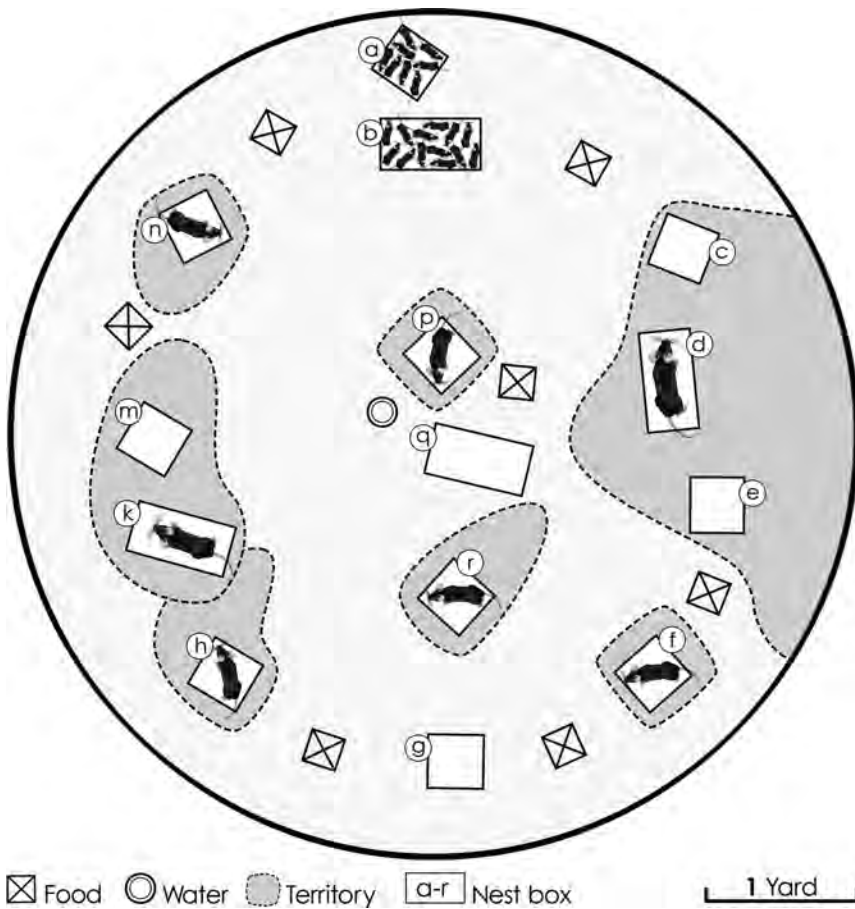


Figure 12.7. Skewed distribution of territories in a mouse town. The great majority of available space were owned by a few mice (one male each in box d, f, h, k, n, p, r), whereas the majority of mice were crowded into two boxes (a: eight males and b: thirteen males and nineteen females—only males are shown). Boxes c, e, g, m, and q were unoccupied but patrolled by territorial males. The size of the mouse indicates its magnitude of dominance in the social hierarchy.

thirteen males and nineteen females were crowded into a single house. There were no nests in the unoccupied boxes, and the territorial males did not allow other mice to settle in them. The territorial males defended their specific boxes against all comers. This social structure persisted for months.

The males' territories were not confined to the boxes alone. Large floor areas also belonged to certain males. The invisible boundaries between their respective domains were nevertheless very real and recognizable to all mouse citizens. The most aggressively dominant mouse lived in a large nest-box (d), and

he spent most of his waking hours patrolling his territory or guarding over it from a vantage point at one corner of the top of his home box. “Here he would crouch, sniffing in the air, and peering about below, and if a strange mouse entered its territory, he would hurl himself through the air, like a tiger onto the intruder and drive it away.”⁶⁰ His attacks were invariably successful, and afterward he always made the rounds of his territory.

The eight males that lived together in box (a) without any nest material were the weakest and most downtrodden of the colony. Box (b) housed two-thirds of the colony’s population. This box was dominated by a single male, which constantly tyrannized most other members of his house, although he rarely challenged the status of the seven males who guarded private houses. These “slum” mice rarely mated, and most offspring derived from the territorial males.

Extrapolation to Human Interactions

Compare these observations on mice with the statistics from another species, *Homo sapiens*. The wealthiest 10% of households own half of the total wealth in the United States and many industrialized countries. Almost all of rest is distributed among half of the population, while the poorest 40% of people own as little as 3% of the wealth. Income distribution is well described by a log-normal distribution, which allows for an unambiguous quantification of inequality.⁶¹ The socioeconomic difference tends to grow over time, transmitting advantages to the minority and disadvantages to the majority from one generation to the next. In the 1980s, the average disposable income of the richest 10% in Europe was seven times higher than that of the poorest 10%; today, it is almost ten times higher. This “Great Divide” not only segregates the rich and the poor in terms of incomes and accumulated wealth but also affects education, access to jobs, well-being, and life expectancy.⁶²

60. Crowcroft’s book (1966) is perhaps the best and most entertaining I ever read on mouse behavior. It contains careful observations of the social and territorial behavior of the house mouse (*Mus musculus musculus*), with lots of detailed and remarkable observations. In repeated experiments, it remained typical that two-thirds or more of the mice were crowded into one shelter, independent of the size of the pen or whether food was scarce or abundant.

61. Vilfredo Pareto (see Koch, 1998) observed that 20% of the population owned about 80% of the land in Europe. This came to be known as the 80–20 law. However, today, the richest 10% owns more than 85% of world wealth. Pareto’s power law is only applicable to the tail of income distribution. Incomes follow a log-normal distribution, not power laws (Glomm and Ravikumar, 1992; Deininger and Squire, 1996; Benabou, 2000).

62. Rising economic inequality can also lower social trust in institutions, increase gaps in education, generate a perception of injustice, lead to intolerance and discrimination, and fuel

Like the “rich club” of highly active neurons with better connections, more effective access to information, and the collective ability to make quick decisions, human group formation may be guided by similar rules. Well-disciplined armies and police controlled by an organized elite have always effectively dominated large but disorganized masses of people. At the turn of the twentieth century, 3 million noblemen and their officials lorded over 180 million peasants and workers in Russia. Then along came the super well-organized Communist Party, with just 23,000 members in 1917, which gained control over the vast Russian Empire overnight and eventually dominated one-sixth of the world’s population. As one of the “beneficiaries” of this system, I had a hard time understanding how such a small fraction of people can maintain such powerful authority. Now I understand it better: organization is the key.⁶³ Paradoxically, all their social injustice was done under the banner of equality.

I began this chapter by explaining that skewed distributions offer stability when many competing factors are in play, which is the case in biology in general and in the brain specifically. This principle may apply to human social groups as well. Egypt with its pharaohs existed for more than three millennia. India’s caste system is among the world’s oldest surviving social hierarchies.⁶⁴ The long-lived Chinese dynasties, the Holy Roman Empire, the Khmer Empire, and the Mayan civilization were all highly stratified systems that bestowed many privileges on the upper classes while sanctioning repression of the lower ones by the privileged minority. In contrast, Greek democracy and communist “democracies” (or the “dictatorship of the majority”) were ephemeral (even if it did not feel that way for many). The moment the Soviet grip on these countries was lifted, the artificially maintained wealth distribution shifted back to the skewed form in less than a decade.

Can we learn anything from the similar distribution of owned stuff in species as different as mice and humans? Some economists argue that artificially increasing the income of the poor alone will not produce the desired impact. I agree. It is not only the magnitude of wealth differences that matters, but also how people feel about it. Inequality is relational at both economic and emotional levels. In addition to compressing the left end of the log-normal distribution,

political instability. The social contract varies considerably across nations. For supporting statistics, see Centre for Opportunity and Equality, <http://oe.cd/cope-divide-europe-2017>.

63. The emphasis on organization was the key point of Yuval Harari’s recent book *Homo Deus* (2017).

64. The division of labor creates specialized behavioral groups and occurs frequently in communal animal societies, including some mammals. Group membership determines lifestyle and heredity of privileges. In contrast, the mainly hunter-gatherer societies of Aboriginal Australians have existed for 60,000 years, perhaps because they live in small segregated communities.

we must compress the right end as well. We need a societal version of the homeostatic compensatory mechanism of non-REM sleep to combat the intrinsic drive of technological development and global economy toward widening the gap between the two tails (Chapter 9). Unfortunately, history shows that such compensatory mechanisms are exceptionally rare. Inequality is usually reduced suddenly only by natural disasters, plagues, wars, and revolutions.⁶⁵

Skewed distributions in human societies may be regarded as an “is-ought problem,” the problem of identification of natural properties with moral properties.⁶⁶ Just because things are as they are, it does not mean that they ought to stay that way. Our social contract, which has matured over hundreds of thousand years by externalizing brain function, should be different from that of the rodents. We have the ability to introduce rules and enforce them wisely so that they counter the forceful tendencies of skewed distributions. Promulgating the idea that we are all born with equal abilities, as the Marxist doctrine naïvely and consistently did, will never lead to social justice or equal opportunities. On the other hand, acknowledging that things are the way they are but do not have to stay that way in human societies is what we should strive for. Unless this idea is understood and acted upon, market economies in their present form will only continue to reinforce social injustice described by log-normal distribution.

Neuronal organization driven by skewed distribution is remarkably different from the expectation of the outside-in framework and is more compatible with a preconfigured yet flexible brain, which is the topic of the last chapter.

SUMMARY

Unlike physics, neuroscience has few generalizable rules that cut across multiple domains of brain organization. The Weber-Fechner law, established in human psychophysics, is an exception. The law states that for any sensory modality, perceptual intensity is a logarithmic function of physical intensity. Therefore, as stimulus strength multiplies, the strength of perception only adds. Distance perception, time perception, and reaction time also vary logarithmically with the distance and time interval, respectively. Error accumulation in short-term memory, and other phenomena also obey this law. In short,

65. Several excellent recent books are available on this important topic. Perhaps not unexpectedly, they differ extensively in their analyses and recommendations (Atkinson, 2015; Payne, 2017; Scheidel, 2017). A concise summary is by Reeves (2017).

66. The “is-ought problem” was most prominently articulated by David Hume in his *A Treatise of Human Nature* (Stanford Encyclopedia of Philosophy; <https://plato.stanford.edu/entries/hume-moral/>).

a variety of sensory and cognitive domains may be governed by similar neuronal mechanisms.

I discussed the hypothesis that the strongly skewed nature of our perceptions and memory result from log-normal distributions of anatomical connectivity at both micro- and mesoscales, synaptic weight distributions, firing rates, and neuronal population activity. Nearly all anatomical and physiological features of the brain are part of a continuous but wide distribution, typically obeying a log-normal rule. This rule implies that the interactions that give rise to this distribution involve multiplication or division of random factors, resulting in values that can span several orders of magnitude. Neuronal networks with such broad distributions are needed to maintain stability against competing needs, including wide dynamic range, redundancy, resilience, homeostasis, and plasticity.

Neurons on the two ends of the log-normal distribution of activity organize themselves differently. Fast-firing neurons are better connected with each other and burst more than slow-firing neurons. The more strongly connected faster firing neurons form a “rich club” with better access to the entire neuronal population, share such information among themselves, and, therefore, generalize across situations. In contrast, slow firing neurons keep their independent solitude and elevate their activity only in unique situations. The two tails of the distribution are maintained by a homeostatic process during non-REM sleep. The emerging picture is that a simple measure, such as the baseline firing frequency, can reveal a lot about a neuron’s role in computation and its wiring properties.

The log-normal distribution rule acting at so many different levels of neuronal organization allows the brain to make relational judgments and, in humans with language, to discretize the extreme ends of such distributions with words. The left and right ends of the log distribution appear to convey different or opposing effects, as if they were discrete groups with distinct features.⁶⁷ In reality, though, they are part of a continuum with long tails. An ever-active minority of neurons at the right end may be responsible for generalizing across environments, which enables the brain to regard no situation as completely unknown. On the other hand, the majority of less-active neurons can be mobilized to precisely distinguish one situation from another and label each as distinct.

67. This may explain the philosopher Georg W. F. Hegel’s mysterious law of the transformation of quantity into quality and vice versa. Hegel argued that qualitative differences in nature arise from either differences of composition or differences in quantities of energy. From the point of view of log-normal distribution, this point appears almost obvious.

Neuronal systems organized according to these principles may form brain networks with opposing but complementary features, producing good-enough and fast decisions in most situations using only a subset of the brain's resources. In contrast, extensive brain networks are needed to learn important details.

